

ABRAR FAHYAZ

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Education

University of Dhaka

B.Sc in Computer Science & Engineering

2022 - 2027

CGPA: 3.83/4.00 (4th out of 65)

Technical Skills

Languages: C/C++, Python, Java, Swift, Rust, ARM Assembly;
Frontend: HTML/CSS, TypeScript, Vue, React, NextJS, JavaFX;
Backend: NodeJS, Django;
ML: PyTorch, Tensorflow, Keras, spaCy, Scikit-learn;
Tools: Qt, Git, Docker, Railway, Vercel;
Agentic AI: Playwright, LangGraph, MCP;

Work Experience

Fellowship.AI

September 2025 – November 2025

Machine Learning Fellow

California, United States (Remote)

- Cohort 35 Fellow of the Launchpad.ai fellowship program.
- Worked on a Real Estate Listings Optimization project with 6 other international fellows, where we designed and implemented a Supabase database to manage realtor, property, location, and analysis metadata.
- After experimenting with multiple scraping strategies, we developed a robust data pipeline for parsing, cleaning, and uploading scraped listings at scale.

Scale AI, Inc

September 2024 – August 2025

Advanced AI Coder

California, United States (Remote)

- Helped in the training and optimization of LLMs in generative coding through code validation and debugging in Swift, Python, JS and C/C++.
- On Outlier.ai, worked on over 10 projects as an AI Trainer (Coding), and reviewed tasks from fellow trainers as a Reviewer.
- Further worked on the OpenAI's Feather program on advanced GPT training (*Hard Colosseum* and *Colosseum V2*).

GenMorphics AI Solution

January 2024 – October 2024

Quality Assurance Analyst

Dhaka, Bangladesh (Remote)

- Worked as a Quality Assurance Analyst for fine-tuning an LLM at SuperAnnotate's Project X and OctopusMath.
- Performed quality assurance on data annotations for LLM training, ensuring accuracy and relevance for end-users.

Udvash-Ummesh

August 2022 – July 2024

Academic Materials Contributor

Dhaka, Bangladesh

- Contributed to question sets for University Admission and Board Examinations.
- Developed educational materials and practice problems for STEM subjects.

Selected Projects

PulseTrace

[Source Code](#)

- Agentic MCP-enabled AI agent capable of scraping social media, and summarizing sentiment through FAISS + BM25 with rerankers, LangGraph, and a self-corrective RAG pipeline. Optional PDF briefing and AI corpus integrated with application following BYOK.
- Beats Last30Days on **nDCG@5** by 38%, and on **precision@5** and **mean norm** by roughly 14%.

P2P Video Streaming Platform

[Source Code](#)

- Decentralized Peer-to-Peer video streaming and storage platform using a custom Distributed Hash Table and direct chunk exchange.
- Instant video playback with sub-linear latency on Streamlit frontend.

Neural Connectome of *C. elegans*

[Source Code](#)

- LIF SNN utilizing numpy and OpenWorm datasets, mapping the entire nervous system as a functional neural network by simulating chemo-taxis.
- All 302 neurons of a hermaphrodite *C. elegans* specimen, alongside over 5100+ chemical and gap synapses implemented.

ChaOS: Turing-Complete OS Emulator

[Video Demo](#) — [Source Code](#)

- Fully functional OS with virtual memory, process scheduling, I/O handling, filesystem and shell layers.
- Custom ISA with compiler toolchain for batch scripting allow Turing completeness.

Theia: Android Photo Sharing Platform

[Video Demo](#) — [Source Code](#)

- Android application for photo sharing with advanced features including real-time image capture, multi-layer editing (filters, annotations), and cloud-based storage with Firebase.
- In-built facial recognition using ML Kit and location clustering via Geo-tagging, and optimized data synchronization through real-time database.

Strategic War Simulator: Military Visualization Suite

[Video Demo](#) — [Source Code](#)

- Military simulation platform for dynamic troop movement visualization and battle timelapses.
- Integrated a brigade commander database with 100+ historical units and Commanders, featured in [The Daily Star](#).

Research Experience

Evidential Spiking Neural Network Framework for Alzheimer's Dementia Classification

2026

Ongoing

- Working on a novel Dempster-Shafer Evidential SNN framework for AD diagnosis and rehabilitation under Dr. Maybin K. Muyeba.
- Work tackles multi-modal uncertainty using Dempster-Shafer Theory, integrated with a Spike-former architecture, for explainable AD diagnosis.
- The research project is being done under the Salford Institute for Dementia and Ageing.

Neuroplastic Neural Networks: Structural Plasticity and Hebbian Updates

2026

Springer Nature Computer Science, Volume 7, Article 474

[Show Publication](#)

- First Author of extended study on the eponymous conference paper, supervised by Dr. Kaveh Kiani and Dr. Maybin K. Muyeba (*UofSalford*), who were judges at the ICDSAIA 2025.
- Includes extended theoretical foundation, statistical validation and ablation, detailed literature review, and a CNN framework.
- Underwent two rounds of peer-review, selected for publication in SNCS (Q2) on March 2026, published in May 2026.

Neural Networks with Neuroplasticity: Connection Pruning and Hebbian Updates

2025

ICDSAIA 2025, Part II, CCIS Vol. 2682, Ch. 5, pp 60-74

[Show Publication](#)

- Preliminary research into neuromorphic neural network architectures with adaptive learning capabilities, built on the conceptual framework of Dr. Donald Hebb.
- Conference paper presented at the International Conference on Data Science, AI and Applications (ICDSAIA 2025). Published as a book chapter in CCIS (Q4).

Achievements

- **1st Place** in multiple Inter-University CTFs: NRFCTF 2024, CCAF Cyber Hackathon 2024, CyberSiege 2025, BUPCTF 2025
- **3rd Place** in DUCTF 2023, **4th Place** in IUTCTF 2023, BUETCTF 2024, and **5th Place** in CyberRaid 2025
- **7th Place** in IUT Code Refactoring Contest 2023
- **2nd Place** in Poster Presentation at TechTron 2025 — Topic: *SinGAN*
- **4th Place** in Three Minute Thesis - Presentation (TMT-P) — Topic: *NLP*
- **Finalist** (*top 26*) in the 1st DURS Undergraduate Research Summit 2023 — Topic: *NLP*
- **Finalist** in SUST CSE Carnival 2024 and Infinity AI Buildfest 2026 — Field: *Hackathon*